

KRILL OIL SUPPLEMENTATION INCREASES THE OMEGA-3 INDEX WITHOUT ALTERING BODY COMPOSITION IN YOUNG, UNTRAINED ADULTS: A RANDOMISED CONTROLLED TRIAL

Elham Yaghoobi¹, Giselle L. Allsopp¹, Rhiannon M.J. Snipe¹, Chris S. Shaw¹, Aaron Fox¹, Gunveen Kaur¹, Angus M. Hunter², and Lee Hamilton¹

¹ Institute for Physical Activity and Nutrition, Deakin University

² Department of Sport Science, Nottingham Trent University

INTRODUCTION

Krill oil (KO) is a rich source of long-chain omega-3 polyunsaturated fatty acids (LC n-3 PUFAs) in a phospholipid-bound form and contains the antioxidant astaxanthin, which may enhance bioavailability and modulate physiological responses (1). LC n-3 PUFAs may influence lipid metabolism and muscle protein synthesis, potentially affecting body composition (2, 3). This investigation aimed to determine the effects of 4 g/day KO supplementation on body composition, including fat mass, lean mass, bone mineral density (BMD), and visceral adipose tissue (VAT), and the Omega-3 Index (O3I; defined as %EPA + DHA of total erythrocyte fatty acids) in young, untrained adults.

METHODS

Thirty healthy, untrained adults (males and females; age <40 years; BMI <30 kg/m²) were enrolled in a double-blind, placebo-controlled trial between July 2024 and October 2025. Participants were randomised to receive either 4 g/day KO (n = 15) or a mixed vegetable oil placebo (PLA; n = 15) for ≥8 weeks. Body composition was assessed via dual-energy X-ray absorptiometry (DXA) at baseline and post-supplementation. Finger-prick blood samples were collected concurrently to analyse whole-blood fatty acid profiles and the derived O3I.

RESULTS AND DISCUSSION

There was no significant interaction or main effect of KO supplementation on body composition parameters (all $P > 0.05$); these findings remained consistent following sensitivity analyses. At baseline, O3I did not differ significantly between groups (KO: $4.52 \pm 0.42\%$; PLA: $4.65 \pm 0.51\%$). Following the 8-week supplementation period, the O3I increased significantly in the KO group compared to the PLA group (KO: $6.67 \pm 1.73\%$; PLA: $4.85 \pm 0.46\%$; $P < 0.001$).

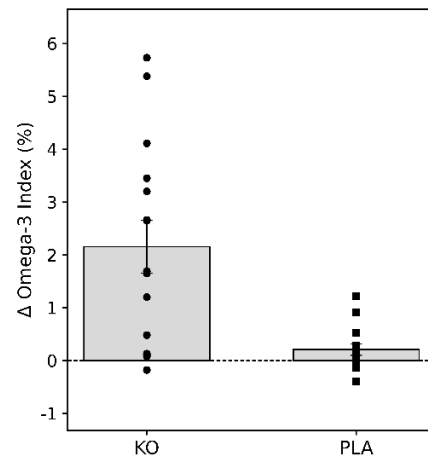


Figure 1. Changes in the Omega-3 Index in KO and PLA groups at pre- and post-supplementation. Bars = mean $\pm$ SEM; points = individual Δ (post - pre). Abbreviations; KO: krill oil, PLA: placebo.

CONCLUSIONS

A minimum of 8 weeks of 4 g/day KO supplementation increased the O3I and shifted the dried blood spot fatty-acid profile toward higher n-3 status in healthy untrained adults without meaningfully altering body composition.

ACKNOWLEDGEMENTS

The authors gratefully acknowledge Deakin university for supporting this project.

REFERENCES

1. Loukil I, Vachon A, Çaku A, Plourde M. Krill oil increases plasma omega-3 fatty acids more than fish oil in healthy adults: a double-blind randomized controlled trial. *Am J Clin Nutr.* 2026;124(1):101346.
2. Moradi S, Alivand M, KhajeBishak Y, AsghariJafarabadi M, Alipour M, Chilibeck PD, et al. The effect of omega3 fatty acid supplementation on PPAR γ and UCP2 expressions, resting energy expenditure, and appetite in athletes. *BMC Sports Sci Med Rehabil.* 2021;13(1):48.
3. Smith GI, Atherton P, Reeds DN, Mohammed BS, Rankin D, Rennie MJ, et al. Dietary omega-3 fatty acid supplementation increases the rate of muscle protein synthesis in older adults: a randomized controlled trial. *Am J Clin Nutr.* 2011;93(2):402–12.